

Sreenivasulu Ummadi

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PROFESSIONAL SUMMARY

Full-stack software engineer with 5+ years at Deloitte and Accenture, building HIPAA-aligned, compliance-grade distributed systems for enterprise clients in regulated industries. Deep expertise in event-driven data pipelines with exactly-once semantics, PHI-grade security controls, and AI-driven automation directly aligned with Optum's prior authorization AI and healthcare analytics roadmap. Passionate about engineering that improves health outcomes — reducing manual processes, accelerating care decisions, and making healthcare data accessible, secure, and actionable at scale.

TECHNICAL SKILLS

Languages: Java, Python, C#/.NET Core, JavaScript/TypeScript, SQL, Scala

Frameworks: Spring Boot, ASP.NET Web API, React, Angular, Django, Node.js, xUnit

Cloud & Infrastructure: AWS (EKS, S3, Lambda, EMR, SQS), Azure, Kubernetes, Docker, Terraform, CI/CD

Data & Messaging: Apache Kafka, Apache Spark, Snowflake, PostgreSQL, Oracle, Redis, MongoDB, Elasticsearch

AI & Automation: LangGraph, Multi-Agent Orchestration (aligned with prior auth AI patterns), pgvector, Python ML Pipelines

Healthcare & Compliance: HIPAA-compliant architecture, PHI encryption, audit trails, AES-256/TLS 1.3, OAuth2/JWT, RBAC, OWASP

Architecture: Microservices, Event-Driven, CQRS, Event Sourcing, Exactly-Once Semantics, FHIR-aligned REST APIs

EXPERIENCE

Deloitte

Consultant

Hyderabad

June 2025 – Present

- Implemented HIPAA-aligned defence-in-depth security controls across enterprise data systems — AES-256 encryption at rest, TLS 1.3 in transit, OAuth2/JWT with RBAC for PHI-equivalent access control, OWASP-compliant CI/CD scanning, and immutable OpenTelemetry audit logs — reducing vulnerability exposure by 40% with zero security incidents.
- Built a custom OpenTelemetry structured audit logger producing immutable, tamper-evident access logs for all sensitive data operations — reducing mean-time-to-detect anomalies by 60% and enabling compliance-grade audit trails aligned with HIPAA record-keeping requirements.
- Achieved 95% code coverage through comprehensive unit, integration, and end-to-end tests with chaos engineering and failure injection — ensuring the reliability standards required of healthcare-grade software.
- Pioneered a multi-agent AI pipeline using Python and LangGraph — architecturally aligned with Optum's prior authorization AI and clinical decision support patterns — achieving 94% classification accuracy and reducing manual review time by 35%.
- Designed and deployed an Agentic AI test orchestration system that synthesizes unit test suites and edge-case payloads from code diffs, reducing manual test authorship overhead by 40% while maintaining compliance standards.
- Built an intelligent agentic monitoring assistant that parses system logs, detects anomalous patterns, and surfaces real-time remediation strategies — proactively preventing 3 production incidents, directly protecting end-user access to critical services.
- Accelerated enterprise CI/CD pipeline by authoring Kubernetes deployment manifests, reducing deployment cycle time from 45 minutes to under 8 minutes across 6 feature teams — enabling faster delivery of member-facing healthcare features.
- Optimized API performance by 40% and database query efficiency by 25% through latency profiling, query tuning, and Redis caching — cutting client-facing latency by 800ms at peak load for 50,000+ daily users.

Accenture

Full Stack Engineering Senior Analyst

Hyderabad

March 2022 – June 2025

- Proactively led a full WCAG 2.1 accessibility audit and remediation — remediating 47 issues, achieving 100% compliance, and embedding accessibility checks into the PR pipeline — making the platform accessible to users with disabilities and meeting ADA/Section 508 standards applicable to healthcare software.
- Maintained 99.9% production uptime for enterprise systems serving 100,000+ concurrent users, treating every outage as a direct impact on the people dependent on those systems.

- Resolved 300+ high-priority production defects across distributed environments with zero security incidents — ensuring the reliability and data integrity required of healthcare-grade enterprise software.
- Identified and escalated a PHI-equivalent data exposure risk in a proposed caching design, prepared a formal risk assessment documenting the compliance implications, and implemented a compliant alternative that passed the security audit with zero findings.
- Mentored 5 junior engineers through structured pairing and architecture reviews — 2 promoted within 18 months — building team capability and raising the engineering quality bar for the entire team.
- Led RFC and technical steering across 4 cross-functional teams, cutting feature design phases by 20% through proactive alignment and transparent communication.

Accenture

Bengaluru

Full Stack Engineering Analyst

September 2020 – March 2022

- Reduced cross-device rendering latency by 20% and improved mobile user retention by 15% through scalable Angular (v8+) and TypeScript front-end architectures serving 50,000+ enterprise users.
- Accelerated page load speeds by 35% and reduced data synchronization latency by 30% through API optimizations and high-throughput RESTful integrations.
- Achieved 100% WCAG accessibility compliance and 95% automated test coverage, expanding the accessible user base and eliminating production regressions.

PROJECTS

Distributed Event-Driven Data Pipeline (Claims-Scale Architecture) | *Java, Kafka, Redis, PostgreSQL, Kubernetes*

- Architected a fault-tolerant event-driven data pipeline processing 20,000+ concurrent records per second with exactly-once semantics — structurally equivalent to a healthcare claims processing pipeline — with idempotency keys, Transactional Outbox pattern, and immutable audit logs meeting HIPAA record-keeping standards.
- Designed strict data isolation with RBAC access controls and row-level security, ensuring PHI-equivalent data is accessible only to authorized roles with full audit logging of every access.
- Guaranteed 99.99% data consistency with crash recovery via Kafka event sourcing — no records lost or duplicated across failure scenarios.

Agentic AI Compliance Guardrail Engine (Prior Auth–Aligned) | *Python, LangGraph, pgvector, Kafka, AzureML*

- Engineered an autonomous multi-agent AI pipeline using Python and LangGraph — architecturally identical to Optum’s prior authorization AI pattern (intake → classification → decision → human-in-the-loop escalation) — achieving 94% classification accuracy and 35% reduction in manual review time.
- Built human-in-the-loop escalation with full immutable audit trail of every AI decision — ensuring compliance-grade accountability and enabling reviewers to audit every autonomous action.
- Reduced LLM inferencing cost by 30% through Redis semantic caching, making the system cost-effective at healthcare data scale.

Resilient API Gateway with HIPAA-Aligned Security | *C#/.NET, Redis, Docker, Kubernetes, AWS* May 2026 – Present

- Designed a high-availability API gateway with defence-in-depth security controls — per-client rate limiting, Circuit Breaker, OAuth2/JWT enforcement, and PHI-safe audit logging — enabling zero-downtime migration of legacy endpoints to containerized microservices.
- Implemented stale-cache graceful degradation, maintaining partial service availability during backend outages and meeting 99.9% SLA commitments critical for healthcare member-facing systems.

EDUCATION

Karunya University

Bachelor of Technology in Computer Science and Engineering

Coimbatore, Tamilnadu

2020

CERTIFICATIONS

- AZ-900: Microsoft Azure Fundamentals
- AI-900: Microsoft Azure AI Fundamentals

ACCOMPLISHMENTS

- ACE’22 Performer of the Year Award — Accenture
- ACE’23 Extra Mile Award for Offshore Individual — Accenture
- FY26 Applause Award — Deloitte